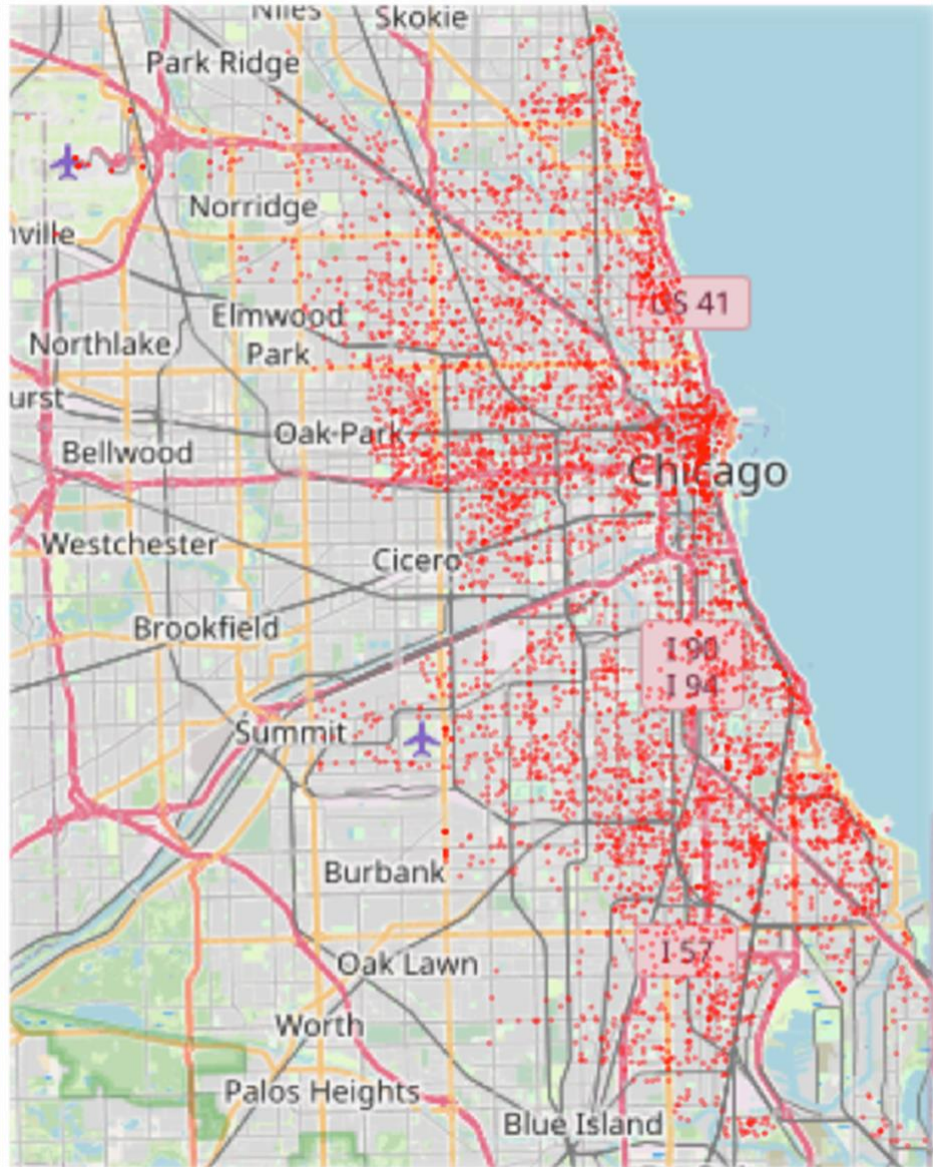


Spatial-Temporal Analysis and Prediction of Urban Crime in Chicago, USA



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Introduction

Urban crime exhibits clear spatial and temporal patterns, making it a key area of analysis for urban policy and law enforcement. This report examines three years of crime incidents in Chicago, focusing on both spatial hotspots and temporal trends. The analysis integrates techniques of exploratory spatial-temporal data analysis with predictive modelling. Crime has long been recognised as unevenly distributed across space and time (Block, 1984). Previous studies highlight the concentration of crime in specific locations and periods, often examined through hotspot mapping and seasonal analysis. Building on this foundation, this study applies spatial clustering techniques alongside forecasting and machine learning approaches to analyse crime patterns. It further evaluates whether machine learning models offer improved predictive performance compared to traditional time series methods, while maintaining interpretability and reproducibility.

The dataset was obtained from the City of Chicago's (2024) public crime. We extracted all reported crimes from January 2022 through December 2024, yielding 753,676 incidents after cleaning. Each record includes the date/time of the offense and the geographic coordinates (latitude/longitude) of the incident, along with attributes like offense type. The data is restricted to valid coordinates within the city bounds and removed any obvious outliers. The cleaned data was then converted into a spatial data frame using the WGS84 projection coordinates. The dataset includes information on the timing and location of each offence, enabling both spatial and temporal analysis. This report examines whether exploratory spatial-temporal analysis and forecasting models can reveal persistent crime patterns in Chicago and whether machine learning improves short-term prediction relative to classical time-series approaches.

Exploratory Spatial-Temporal Analysis

Monthly crime counts reveal clear and consistent seasonal variation. Figure 1 shows the total number of crimes per month from January 2022 to December 2024. Crime levels follow a repeating annual cycle, with peaks occurring during the summer months and troughs in winter, particularly in February. This pattern is consistent across all three years, indicating strong seasonality rather than random fluctuation. While no pronounced long-term trend is observed, intra-annual variation is substantial, suggesting that temporal dynamics are dominated by seasonal effects.

To further examine the underlying structure, STL decomposition was applied to the monthly series (Figure 2).

The decomposition separates the series into trend, seasonal, and remainder components. The seasonal component confirms a stable and repeating annual cycle, while the trend component shows only minor variation over time, indicating that overall crime levels remain relatively stable across the study period. The remainder component is comparatively small, suggesting that most variation in crime counts is explained by seasonal and trend effects rather than random noise.

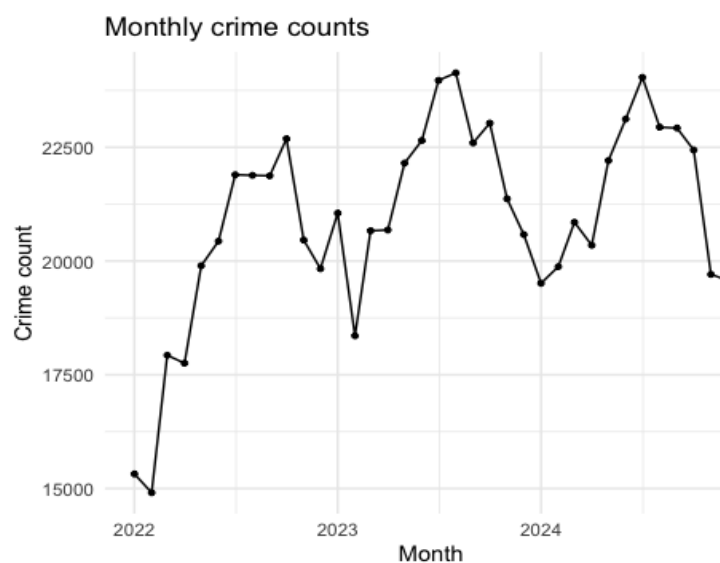


Figure 1. Monthly Crime Counts from 2022-2024 in Chicago.

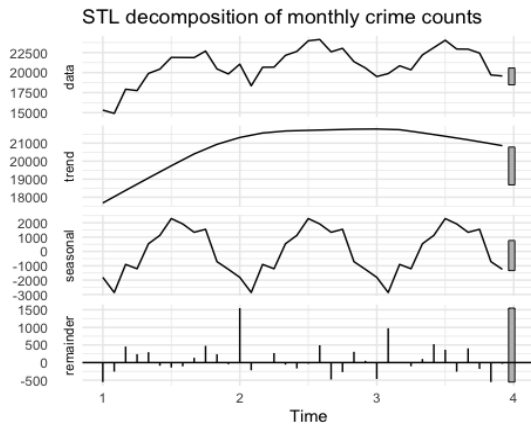


Figure 2. STL decomposition of monthly crime counts.

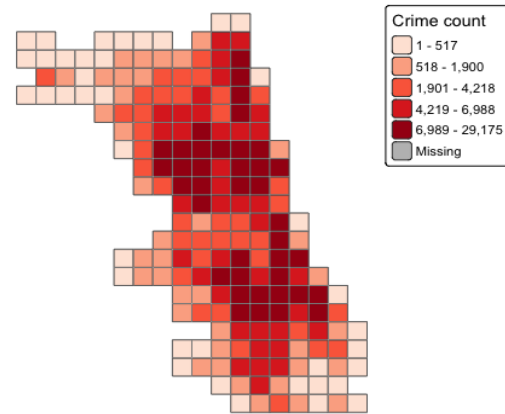


Figure 3. Choropleth map of crime incidents aggregated to a 2km x 2km grid over Chicago. Darker cells indicate higher total crime counts (2022–2024).

The spatial distribution of crime is highly uneven. The choropleth map of crime (Figure 3) counts aggregated to a 2 km grid, showing that crime is concentrated in specific areas rather than evenly distributed across the city. High-density clusters are evident in parts of the West Side, South Side, and central urban areas, while peripheral neighbourhoods experience consistently low crime levels.

This visual pattern suggests strong spatial dependence, which is confirmed through formal spatial autocorrelation analysis. Using a queen contiguity structure, the global Moran's I statistic for grid-level crime counts is 0.65 ($p < 0.0001$), indicating a high degree of positive spatial autocorrelation. This implies that areas with high crime are surrounded by similarly high-crime areas, and low-crime areas cluster together, rather than crime being randomly distributed in space.

Temporal variation is also evident within the weekly cycle. Figure 4 shows crime incidence by hour of day and day of week, revealing higher concentrations during evening periods and on weekends, particularly Friday and Saturday. In contrast, early morning hours and weekdays show lower levels of activity. This pattern suggests that crime is shaped not only by annual seasonality but also by recurring short-term rhythms linked to daily and weekly urban activity.

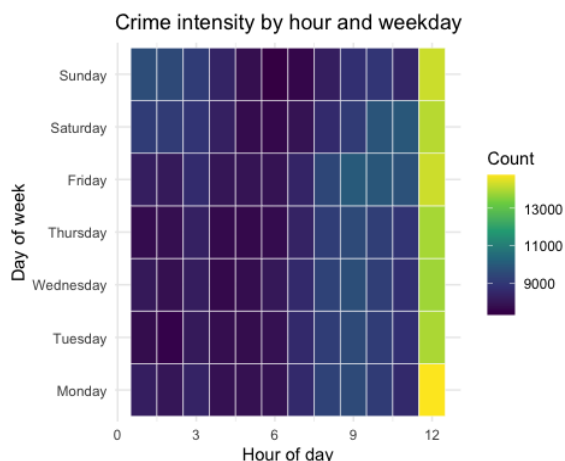


Figure 4. Heatmap of crime incidence by day of week/hour of day.

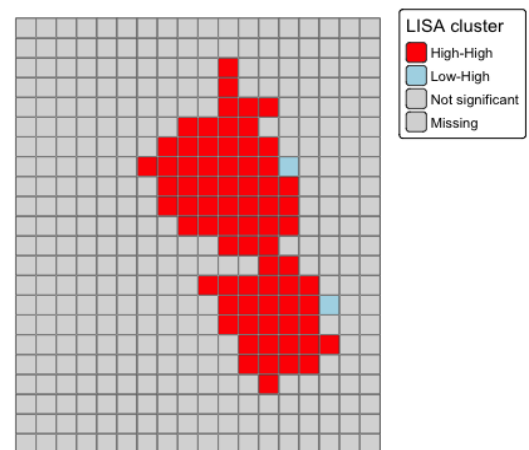


Figure 5. LISA Cluster Map

While Moran's I confirms overall spatial clustering across the city, it does not show where those clusters are located. To identify the specific locations of local hotspots and cold spots, Local Indicators of Spatial Association were computed (Figure 5). To identify local clustering patterns, Local Indicators of Spatial Association (LISA) were computed (Anselin 2024). High-High clusters indicate local hotspots where high-

crime cells are surrounded by similarly high-crime areas, while Low-Low clusters represent cold spots of consistently low crime. The LISA map identifies several significant hotspots in Chicago's central and south-western regions, aligning with known high-crime areas. Similarly, extensive Low-Low clusters appear in the peripheral areas and some upscale sections of the North Side. A few scattered High-Low and Low-High outliers exist, but they are relatively rare. The clustering pattern confirms the hotspot map: crime is geographically concentrated in specific areas of the city, consistent with the criminological principle that a small proportion of locations account for a disproportionate share of incidents (Weisburd 2015).

To assess whether spatial patterns change over time, hotspot maps were generated separately for each year (Figure 6). The results show a high degree of persistence in crime distribution. Areas identified as hotspots in 2022 remain hotspots in 2023 and 2024, and low-crime areas remain consistently low. While there are minor fluctuations in intensity, such as slight increases in 2023 and reductions in certain areas in 2024, the overall spatial structure remains stable.

This persistence suggests that underlying structural factors, such as socio-economic conditions and land use, continue to shape crime patterns over time. The stability of hotspots supports the use of aggregated city-level forecasting, as spatial patterns are unlikely to shift dramatically in the short term.

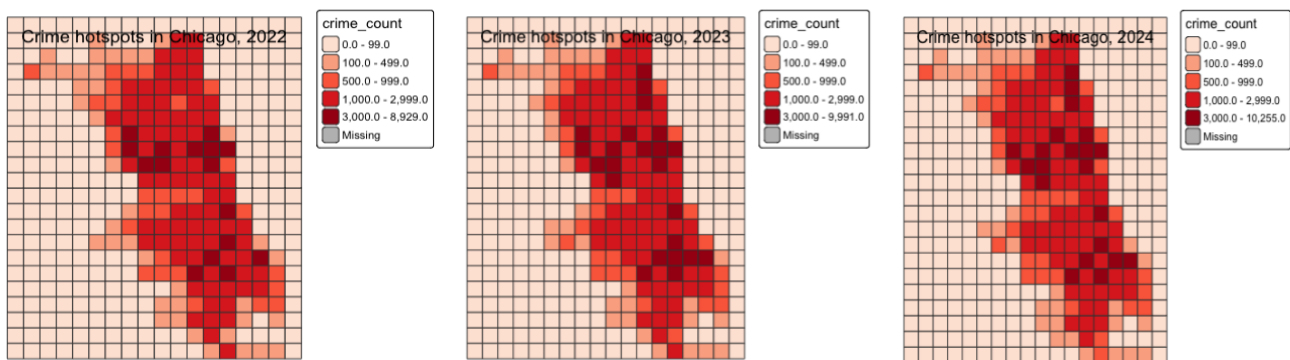


Figure 6. Hotspot map of Crime counts in Chicago 2022, 2023 and 2024.

The exploratory analysis demonstrates that crime in Chicago exhibits strong temporal regularity and spatial clustering. Crime peaks during summer months and weekend evenings, while remaining concentrated in specific urban areas with persistent hotspot structures. These findings directly inform the modelling strategy by showing stable seasonality, short-term temporal regularity, and persistent spatial clustering, which support the inclusion of seasonal forecasting components and the assumption of relatively stable short-term spatial patterns.

Methodology and Results

Before constructing predictive models, it is necessary to consider whether the temporal patterns observed in the data are stable and suitable for forecasting. The exploratory analysis demonstrated strong seasonal regularity in crime counts, alongside short-term temporal dependence. Monthly aggregation was used to establish long-run seasonality, while weekly aggregation was adopted for forecasting because it preserves more observations and captures short-term variation more effectively.

Crime incidents were aggregated into a weekly time series for the entire city. This level of aggregation reduces daily noise while preserving meaningful temporal variation, particularly annual seasonality. Let y_t denote the total number of crimes in week t , resulting in a time series of approximately 156 observations over the period 2022–2024. To evaluate model performance, the dataset was divided into training and testing subsets. The first 80% of observations were used for model estimation, while the remaining 20% were reserved for out-of-sample evaluation. This corresponds to forecasting approximately 30 weeks

ahead, which is appropriate for short-term planning applications. Stationarity is a key requirement for time-series modelling. An Augmented Dickey–Fuller (ADF) test was applied to the weekly series, indicating non-stationarity ($p = 0.53$). First-order differencing was therefore applied, after which the series became stationary ($p \approx 0.01$). The autocorrelation structure of the differenced series was examined using ACF and PACF plots (Figure 7), which informed model specification.

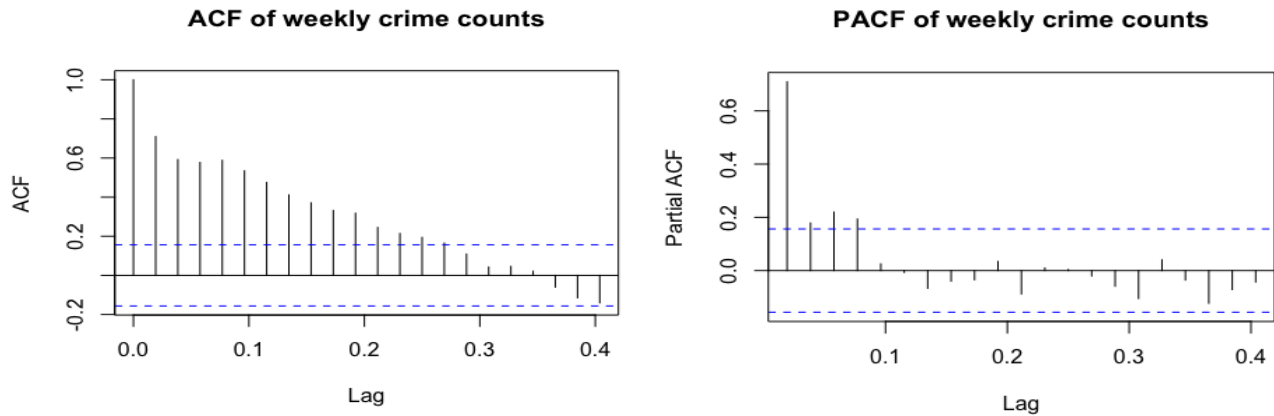


Figure 7. ACF (left) and PACF (right) plots of the differenced weekly crime series.

Four models were implemented to capture different aspects of temporal dynamics:

- **Naive baseline**

A simple persistence model was used as a benchmark, where forecasts are equal to the previous week's observed value.

- **ARIMA model**

A non-seasonal ARIMA(3,1,2) model was fitted to the differenced series. This specification captures short-term temporal dependence through autoregressive and moving average components (Ho and Xie 1998). Residual diagnostics indicated no remaining autocorrelation (Ljung–Box $p = 0.53$), suggesting an adequate model fit.

- **Seasonal ARIMA (SARIMA) (Pfeifer and Deutsch 1980)**

To account for annual periodicity, a SARIMA(1,1,1)(0,1,0)[52] model was implemented. Seasonal differencing at lag 52 captures the yearly cycle observed in the data. Residual tests (Ljung–Box $p = 0.41$) confirm that the model sufficiently accounts for temporal structure.

- **Random Forest model**

A machine learning approach was employed using engineered temporal features, including lagged crime counts (lags 1, 2, 3, 4, 12, and 52), rolling averages, and cyclic seasonal encodings (sine and cosine transformations of week-of-year) (Zhang et al. 2022). The model explained approximately 59% of variance in the training data and allows for flexible, non-linear relationships.

The results are summarised in Table 1 and visualised in Figure 5, which highlights differences in model performance across error metrics. The naive model produced the highest error, indicating that simple persistence fails to capture underlying temporal structure. The ARIMA model improved predictive accuracy, but its lack of seasonal components limited performance during peak periods (Kane et al. 2014; Zhang et al. 2022). The SARIMA model significantly reduced error, demonstrating the importance of incorporating seasonal dynamics. The Random Forest model achieved the best, outperforming all statistical models by capturing both seasonal patterns and short-term fluctuations.

Figure 8 and Figure 9 compare predicted and observed values across models. The Random Forest model more closely follows observed crime levels, particularly during periods of rapid change, indicating its ability

to adapt to complex temporal dynamics. The Random Forest model more closely follows observed fluctuations, particularly during peak periods, compared to ARIMA-based models (Figure 10).

Model	RMSE	MAE
Naive	737	660
ARIMA	584	521
SARIMA	337	264
Random Forest	261	233

Table 1. Summary of models' performances.

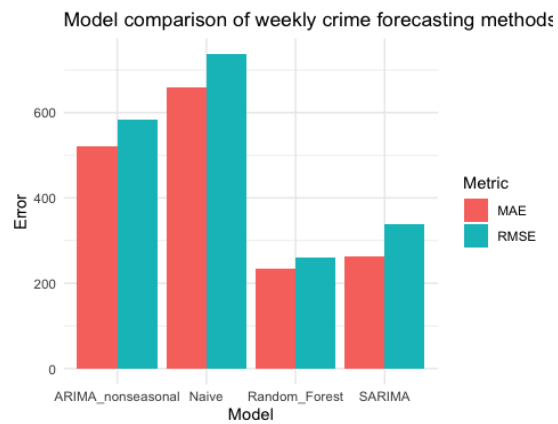


Figure 8. Model comparison bar chart (RMSE & MAE).

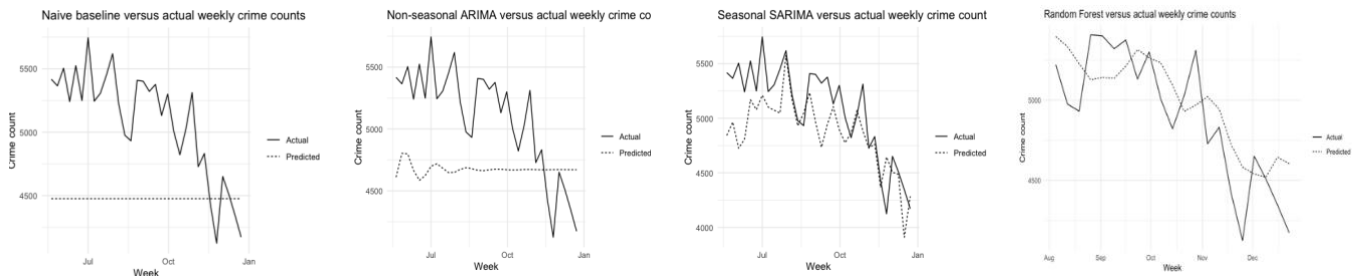


Figure 9. the actual vs. predicted weekly crime counts for ARIMA, SARIMA, and Random Forest models.

To extend the analysis beyond aggregate temporal modelling, spatial information was incorporated through a spatial lag variable. Crime counts were aggregated to grid-level units, and neighbouring relationships were used to construct a spatial lag feature representing local spillover effects. This spatial-temporal Random Forest model achieved an RMSE of 15.73 and an MAE of 9.44 at the grid-week level. These results indicate that predictions deviate by approximately 9–10 crimes per cell per week on average. The inclusion of spatial lag features improves model realism by capturing interactions between neighbouring areas.

However, prediction accuracy varies across space due to the uneven distribution of crime. Low-activity areas exhibit higher relative variability, while high-density areas are more predictable. This highlights the limitations of fine-scale forecasting and suggests that alternative approaches, such as coarser aggregation or hotspot classification, may be more suitable for spatial prediction.

Discussion and Evaluation

The results highlight the importance of modelling both seasonal and short-term temporal dynamics in urban crime data. Among the models tested, Random Forest achieved the highest predictive accuracy, followed by SARIMA, ARIMA, and the naive baseline. This performance hierarchy reflects the extent to which each model captures underlying temporal structure.

The superior performance of the Random Forest model suggests that flexible, data-driven approaches can better capture complex patterns in crime data than traditional linear models. By incorporating multiple lagged features and seasonal encodings, the model effectively learned both annual cycles and short-term dependencies. In contrast, ARIMA-based models are constrained by linear assumptions and predefined

structures. While the SARIMA model successfully captured annual seasonality, it was less responsive to local fluctuations and variation in peak intensity. The improved performance of Random Forest indicates the presence of non-linear effects or interactions that cannot be fully represented in linear time-series models, consistent with findings in previous studies (Rathod et al. 2021).

Despite its accuracy, the Random Forest model is less interpretable than ARIMA-type models (Kane et al. 2014). The SARIMA specification allows direct interpretation of temporal dependencies through its parameters, whereas Random Forest operates as a black box. Feature importance analysis partially addresses this limitation, showing that one-year lag, recent observations, and seasonal indicators are the most influential predictors. This confirms that crime patterns are largely driven by temporal persistence and seasonality. In practice, this presents a trade-off between accuracy and interpretability, where simpler models may be preferred for communication and policy transparency, despite slightly lower predictive performance.

A key limitation of the modelling approach is its reliance on city-level aggregation. While this simplifies the temporal structure and improves predictive stability, it assumes that spatial patterns remain constant over time. The exploratory analysis supports this assumption in the short term, with stable hotspot locations observed across years. However, the models do not capture intra-urban variation, and forecasting at finer spatial scales would likely introduce greater uncertainty due to data sparsity and volatility. More advanced spatial-temporal approaches, such as hierarchical models or spatially structured machine learning methods, could improve localised prediction but were beyond the scope of this study.

The analysis is also limited by the short time and lack of external explanatory variables. Using only three years of data restricts the ability to capture longer-term trends and may reduce model robustness. Additionally, the exclusion of factors such as weather, holidays, or socio-economic conditions means that some variation is treated as unexplained noise. While the results demonstrate that much of the temporal structure can be captured through seasonality and lag effects alone, incorporating exogenous variables could further improve predictive performance and interpretability.

This study does not differentiate between crime types. Aggregating all incidents into a single time series simplifies analysis but masks important differences in underlying dynamics. Different crime categories may exhibit distinct temporal and spatial patterns and modelling them separately could provide more targeted insights for policy and intervention.

Conclusion and Reflection

This study demonstrates that crime in Chicago exhibits strong spatial clustering and predictable temporal patterns. Crime is concentrated in persistent hotspot locations and follows consistent seasonal and weekly cycles (Weisburd 2015). These patterns provide a robust foundation for short-term forecasting.

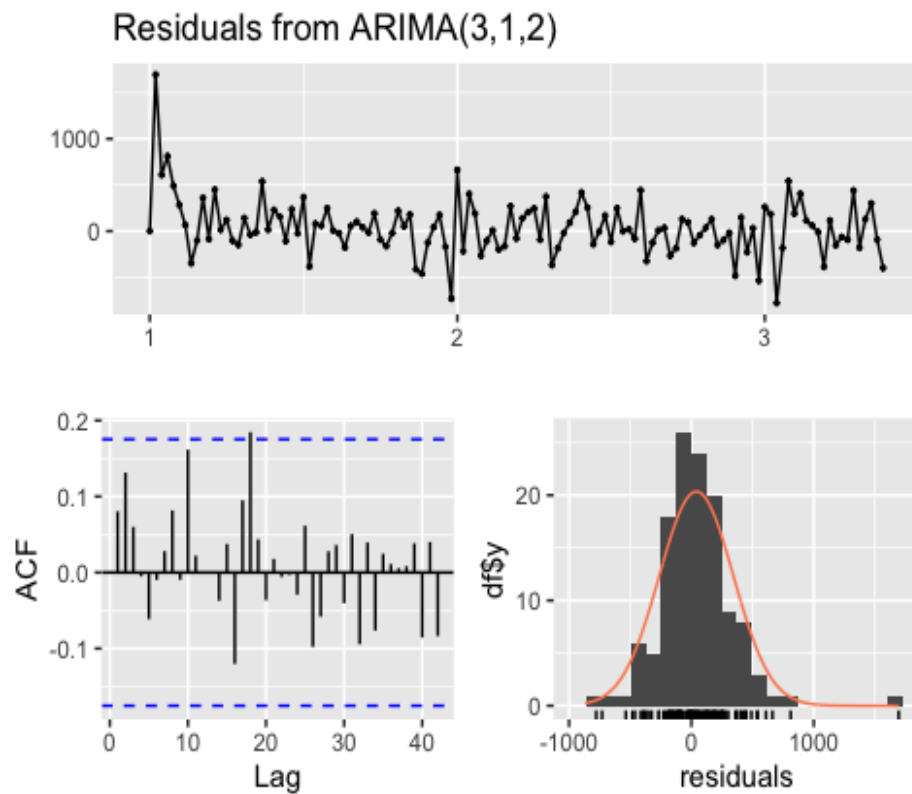
Among the models evaluated, Random Forest achieved the highest accuracy by effectively capturing both seasonal variation and short-term dynamics. The results show that incorporating temporal structure, particularly annual seasonality, is critical for accurate prediction. However, the choice of model involves a trade-off between predictive performance and interpretability, with simpler models such as SARIMA remaining valuable in practice.

From a policy perspective, the findings support targeted and time-sensitive interventions. Persistent hotspots suggest that place-based strategies can be consistently applied, while predictable temporal variation allows for more efficient allocation of resources across seasons and time periods (Block 1984). Forecasting models can further support anticipatory planning by providing short-term projections of crime levels.

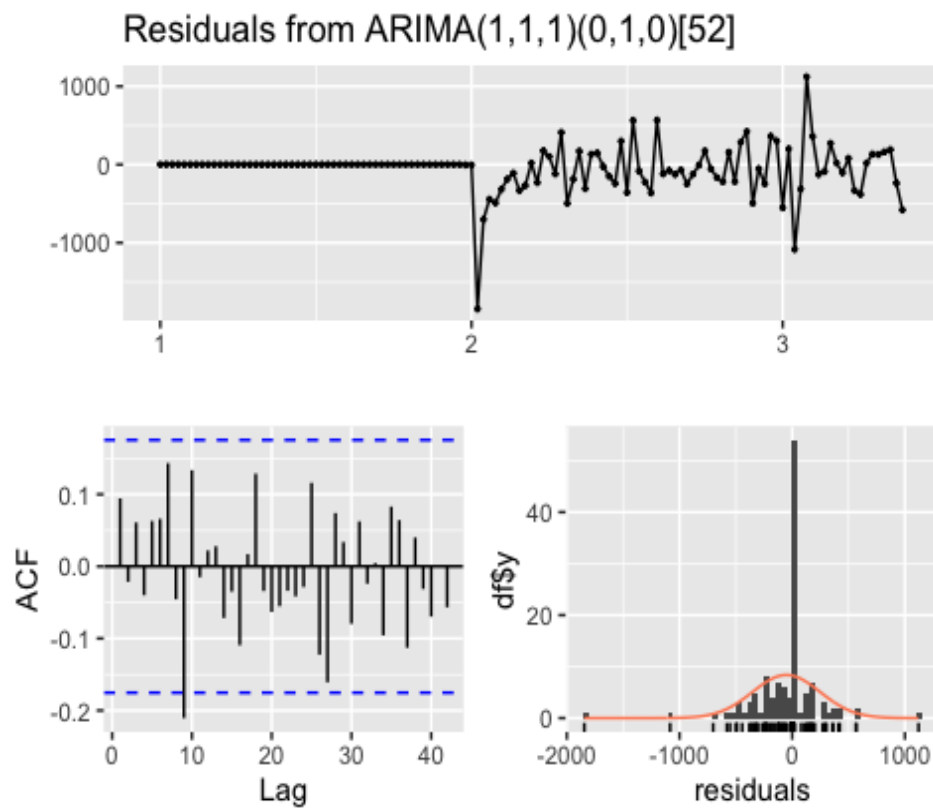
Future work could extend this analysis by incorporating longer time series, external explanatory variables, and finer spatial resolution. Integrating spatial and temporal modelling more fully would improve understanding of local dynamics and enhance the practical applicability of crime forecasting.

Appendix

Figure A1: ARIMA residual diagnostics Figure



A2: SARIMA residual diagnostics Figure



A3: Random Forest feature importance

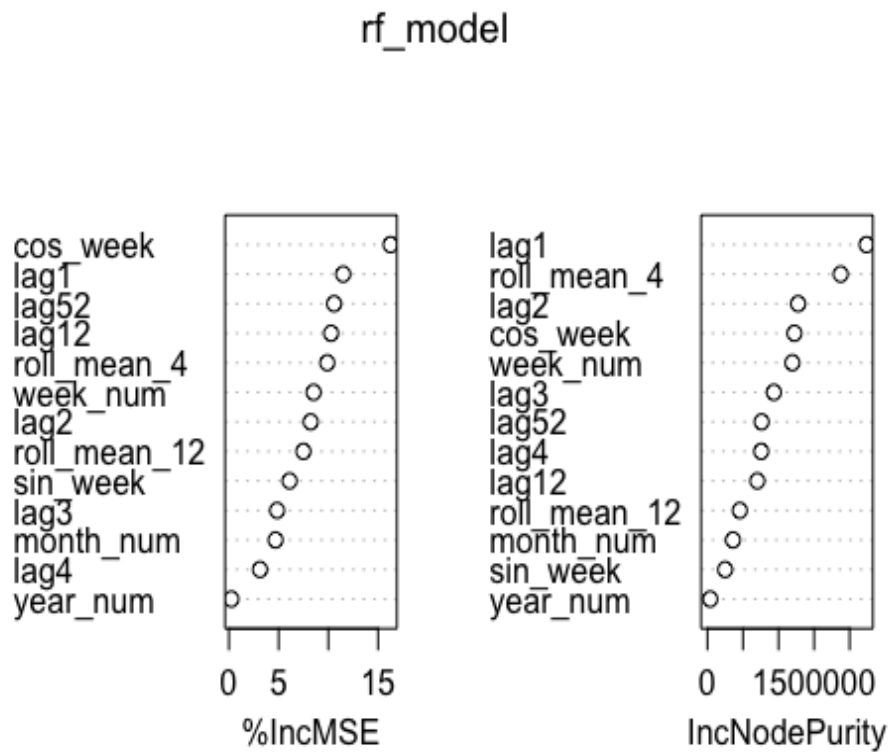


Figure A4: combined actual vs predicted plot for all main models

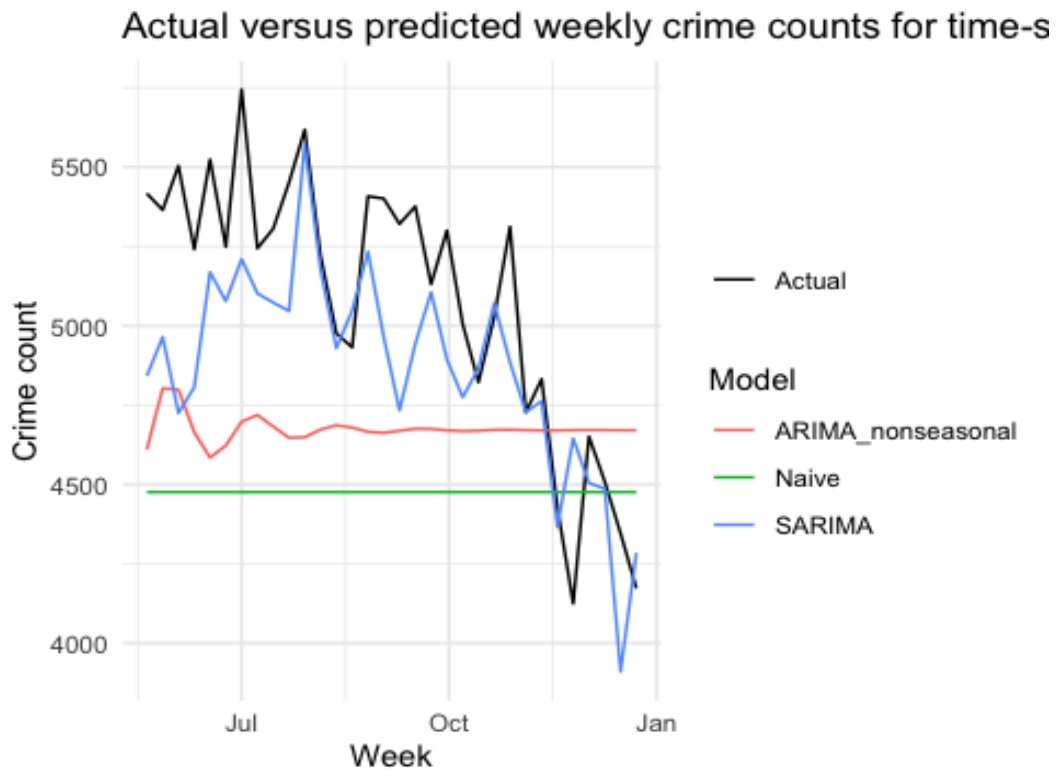


Table A1: ADF test results and Ljung-Box test results in one compact table

Test	Series / Model	Test statistic	Lag / df	p-value	Interpretation
Augmented Dickey-Fuller	Weekly crime series before differencing	-2.1095	Lag order = 5	0.5307	Non-stationary
Augmented Dickey-Fuller	Weekly crime series after first differencing	-6.2001	Lag order = 5	0.0100	Stationary after differencing
Ljung-Box	Residuals from ARIMA(3,1,2)	Q* = 18.915	df = 20	0.5273	No significant residual autocorrelation
Ljung-Box	Residuals from SARIMA(1,1,1)(0,1,0)[52]	Q* = 23.889	df = 23	0.4100	No significant residual autocorrelation

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